

# Chapter 1: Combinatorial Analysis

## STAT2920: Introduction to Probability

Mohamed Belalia, PhD

University of Windsor  
Department of Mathematics and Statistics



University  
of Windsor

- Apply the basic principal of counting, **permutations** and **combinations** for counting the number of different ways that a certain event can occur.
- Recall the basic rules on the set theory.
- Apply the axioms of probability in assigning probabilities to **events** in a **sample space**.
- Utilize proper counting techniques to calculate probabilities for various events when sample space having **equally likely** outcomes.
- Compute **conditional probabilities** using the definition for conditional probability.
- Compute probabilities of certain events using **multiplication rule**.
- Calculate probabilities using **Bayes' theorem** and the **law of total probability**.
- Distinguish the **independent** or **dependent** events using the definition of independent events.
- Recognize conditional probability is a probability, thus have all the properties of probability.

- Calculate **probability mass functions(pmf)** for **discrete random variables**.
- Calculate the **expected value** of a discrete random variable.
- Calculate the expectation of a function of a discrete random variable.
- Calculate the **variance** of a discrete random variable.
- Compute probabilities of certain events by applying the properties of a **binomial** random variable.
- Calculate probabilities and expectation for **Poisson** random variables.
- Calculate the **cumulative distribution functions(cdf)** for discrete random variables.
- Apply the properties of **probability density function(pdf)** to properly define a probability density function.
- Calculate the expected value and variance of a **continuous random variable**.
- Compute probabilities for **Uniform** random variables.
- Calculate probabilities given a **Normal** density.
- Recognize the most common probability distributions family.

- Week 1: Course Overview and Combinatorial Analysis
- Week 2: Probability *I*
- Week 3: Probability *II*
- Week 4: Probability Distributions and Probability Densities *I*
- Week 5: Probability Distributions and Probability Densities *II*
- Week 6: Mathematical Expectation *I*
- Week 7: Mathematical Expectation *II*
- Week 8: Special Probability Distributions *I*
- Week 9: Special Probability Distributions *II*
- Week 10: Special Probability Densities *I*
- Week 11: Special Probability Densities *II*

| Assessment | Date               | Weight |
|------------|--------------------|--------|
| Quiz 1     | Friday, Sep. 26    | 5%     |
| Test 1     | Wednesday, Oct. 6  | 25%    |
| Quiz 2     | Friday, Nov. 7     | 5%     |
| Test 2     | Wednesday, Nov. 19 | 25%    |
| Final Exam | TBA                | 40%    |

Table 1: Weights assigned for determination of the raw final score.

## Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

## Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* [here](#).

## Some useful websites:

- Society of Actuaries, [Exam of probability, sample questions](#).
- Society of Actuaries, [Exam of probability, sample solutions](#).

- 1 Introduction
- 2 The Basic Principle of Counting
- 3 Permutations
- 4 Combinations

## Whole image

- Probability aims to formalize everyday sentences of the type:

“The chance of  $E$  is  $p$ ”,

where  $E$  is some **event** and  $p$  is some **measure** of the likelihood of occurrence of that event.

## Example 1.

- There is **35%** chance of rain,
- Due to *Covid-19*, there is **50%** chance that there will be more inflation.
- I believe with a high probability the Liberal Party will win the upcoming election .
- The probability of success **STAT2920** course is relatively high.



- A communication system is to consist of  $n$  seemingly identical antennas that are to be lined up in a linear order.
- The resulting system will then be able to receive all incoming signals—and will be called functional—as long as no two consecutive antennas are defective.
- If it turns out that exactly  $m$  of the  $n$  antennas are defective, what is the probability that the resulting system will be functional?
- For instance, in the special case where  $n = 4$  and  $m = 2$ , there are 6 possible system configurations, namely,

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 |

where 1 means that the antenna is working and 0 that it is defective.

- Because the resulting system will be functional in the first 3 arrangements and not functional in the remaining 3, it seems reasonable to take  $\frac{3}{6} = \frac{1}{2}$  as the desired probability.
- In the case of general  $n$  and  $m$ , we could compute the probability that the system is functional in a similar fashion.
- That is, we could count the number of configurations that result in the system's being functional and then divide by the total number of all possible configurations.

- From the preceding discussion, we see that it would be useful to have an effective method for counting the number of ways that things can occur.
- In fact, many problems in probability theory can be solved simply by counting the number of different ways that a certain event can occur.
- The mathematical theory of counting is formally known as **combinatorial analysis**.

## Theorem 1 (The basic principle of counting).

*Suppose that two experiments are to be performed. Then if experiment 1 can result in any one of  $m$  possible outcomes and if, for each outcome of experiment 1, there are  $n$  possible outcomes of experiment 2, then together there are  $m \times n$  possible outcomes of the two experiments.*

## Proof.

The basic principle may be proven by enumerating all the possible outcomes of the two experiments; that is,

$$\begin{aligned} &(1, 1), (1, 2), \dots, (1, n) \\ &(2, 1), (2, 2), \dots, (2, n) \\ &\vdots \\ &(m, 1), (m, 2), \dots, (m, n) \end{aligned}$$

where we say that the outcome is  $(i, j)$  if experiment 1 results in its  $i$ th possible outcome and experiment 2 then results in its  $j$ th possible outcome. Hence, the set of possible outcomes consists of  $m$  rows, each containing  $n$  elements. This proves the result.  $\square$

## Example 2.

Suppose that someone wants to go by bus, train, or plane on a week's vacation to one of the five East North Central States. Find the number of different ways in which this can be done.

### Solution:

- The particular state can be chosen in  $m = 5$  ways and the means of transportation can be chosen in  $n = 3$  ways.

## Example 2.

Suppose that someone wants to go by bus, train, or plane on a week's vacation to one of the five East North Central States. Find the number of different ways in which this can be done.

### Solution:

- The particular state can be chosen in  $m = 5$  ways and the means of transportation can be chosen in  $n = 3$  ways.
- Therefore, the trip can be carried out in  $5 \times 3 = 15$  possible ways.

# The basic principle of counting: Tree diagram

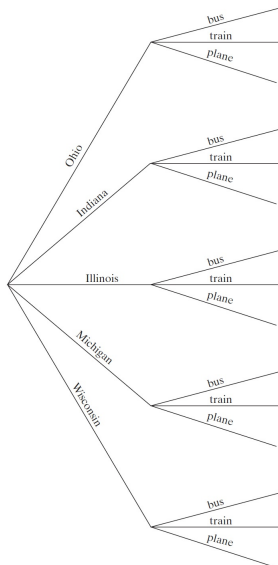


Figure 1: Tree diagram



- If an actual listing of all the possibilities is desirable, a **tree diagram** like that in Figure 1 provides a systematic approach.
- This diagram shows that there are:
  - $m = 5$  branches (possibilities) for the number of states,
  - for each of these branches there are  $n = 3$  branches (possibilities) for the different means of transportation.
- The **15** possible ways of taking the vacation are represented by the **15** distinct paths along the branches of the tree.

## Example 3.

How many possible outcomes are there when we roll a pair of dice, one red and one green?

### Solution:

- The red die can land in any one of six ways, and for each of these six ways the green die can also land in six ways.
- Therefore, the pair of dice can land in  $6 \times 6 = 36$  ways.

## Theorem 2 (The generalized basic principle of counting).

*If  $k$  experiments that are to be performed are such that the first one may result in any of  $n_1$  possible outcomes; and if, for each of these  $n_1$  possible outcomes, there are  $n_2$  possible outcomes of the second experiment; and if, for each of the possible outcomes of the first two experiments, there are  $n_3$  possible outcomes of the third experiment; and so forth, then there is a total of  $n_1 \times n_2 \times \dots \times n_k$  possible outcomes of the  $k$  experiments.*

## Example 4.

A quality control inspector wishes to select a part for inspection from each of four different bins containing 4, 3, 5, and 4 parts, respectively. In how many different ways can she choose the four parts?

### Solution:

- The total number of ways is  $4 \times 3 \times 5 \times 4 = 240$ .

## Example 5.

In how many different ways can one answer all the questions of a true - false test consisting of 20 questions?

### Solution:

- Altogether there are

$$2 \times 2 \times 2 \times 2 \times \dots \times 2 \times 2 = 2^{20} = 1,048,576$$

## Exercise 1.

A college planning committee consists of 3 freshmen, 4 sophomores, 5 juniors, and 2 seniors. A subcommittee of 4, consisting of 1 person from each class, is to be chosen. How many different subcommittees are possible?

## Solution 1.

- We may regard the choice of a subcommittee as the combined outcome of the four separate experiments of choosing a single representative from each of the classes.
- It then follows from the generalized version of the basic principle that there are  $3 \times 4 \times 5 \times 2 = 120$  possible subcommittees.

## Exercise 2.

How many different 7-place license plates are possible if the first 3 places are to be occupied by letters and the final 4 by numbers?

## Solution 2.

By the generalized version of the basic principle, the answer is

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10 = 175,760,000.$$

## Exercise 3.

How many functions defined on  $n$  points are possible if each functional value is either 0 or 1?

## Solution 3.

- Let the points be  $1, 2, \dots, n$ .
- Since  $f(i)$  must be either 0 or 1 for each  $i = 1, 2, \dots, n$ ,
- it follows that there are  $2^n$  possible functions.

- Frequently, we are interested in situations where the outcomes are the different ways in which a group of objects can be ordered or arranged.
- For instance, we might
  - want to know in how many different ways the 24 members of a club can elect a president, a vice president, a treasurer, and a secretary,
  - or we might want to know in how many different ways six persons can be seated around a table.
- Different arrangements like these are called **permutations**.

- How many different ordered arrangements of the letters a, b, and c are possible?
- By direct enumeration we see that there are 6, namely, *abc, acb, bac, bca, cab, and cba*. Each arrangement is known as a **permutation**.
- Thus, there are 6 possible permutations of a set of 3 objects.
- This result could also have been obtained from the basic principle, since the first object in the permutation can be any of the 3, the second object in the permutation can then be chosen from any of the remaining 2, and the third object in the permutation is then the remaining 1.
- Thus, there are  $3 \times 2 \times 1 = 6$  possible **permutations**.



## Definition 1.

A permutation is a distinct arrangement of  $n$  different elements of a set.

- Suppose now that we have  $n$  objects. Reasoning similar to that we have just used for the 3 letters, we get the following result

## Theorem 3 (Permutations).

*There are  $n \times (n - 1) \times (n - 2) \times \dots \times 3 \times 2 \times 1 = n!$  different permutations of the  $n$  objects.*

## Example 1.

In how many different ways can the five starting players of a basketball team be introduced to the public?

### Solution:

- By using Theorem 3, there are  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$  ways in which they can be introduced.

## Example 2.

The number of permutations of the four letters  $a, b, c$ , and  $d$  is 24, but what is the number of permutations if we take only two of the four letters or, as it is usually put, if we take the four letters two at a time?

### Solution:

- We have two positions to fill, with four choices for the first and then three choices for the second.
- Therefore, by Theorem 1, the number of permutations is  $4 \times 3 = 12$ .

## Exercise 4.

A class in probability theory consists of 6 men and 4 women. An examination is given, and the students are ranked according to their performance. Assume that no two students obtain the same score.

1. How many different rankings are possible?
2. If the men are ranked just among themselves and the women just among themselves, how many different rankings are possible?

## Solution 4.

1. Because each ranking corresponds to a particular ordered arrangement of the 10 people, the answer to this part is  $10! = 3,628,800$ .
2. Since there are  $6!$  possible rankings of the men among themselves and  $4!$  possible rankings of the women among themselves, it follows from the basic principle that there are  $(6!)(4!) = (720)(24) = 17,280$  possible rankings in this case.

**Exercise 5.**

Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

**Solution 5.**

- There are  $4! \times 3! \times 2! \times 1!$  arrangements such that the mathematics books are first in line, then the chemistry books, then the history books, and then the language book.
- Similarly, for each possible ordering of the subjects, there are  $4! \times 3! \times 2! \times 1!$  possible arrangements.
- Hence, as there are  $4!$  possible orderings of the subjects, the desired answer is  $4! \times 4! \times 3! \times 2! \times 1! = 6912$ .

- Generalizing the argument that we used in the preceding Theorem, we find that  $n$  distinct objects taken  $r$  at a time, for  $r > 0$ , can be arranged in  $n \times (n - 1) \times \dots \times (n - r + 1)$  ways.
- We denote this product by  $P_r^n$ , and we let  $P_0^n = 1$  by definition.
- Therefore, we can state the following theorem.

## Theorem 4.

*The number of permutations of  $n$  distinct objects taken  $r$  at a time is*

$$P_r^n = \frac{n!}{(n - r)!} \quad (1)$$

*for  $r = 1, 2, \dots, n$*

## Remark 1.

In general,  $n \times (n - 1) \times \dots \times (n - r + 1)$  represents the number of different ways that a group of  $r$  items could be selected from  $n$  items when the order of selection is relevant

## Proof of Theorem 4.

- The formula  $P_r^n = n \times (n-1) \times \dots \times (n-r+1)$  can not be used for  $r=0$ , but we do have

$$P_0^n = \frac{n!}{(n-0)!} = 1.$$

- For  $r=1, 2, \dots, n$ , we have

$$\begin{aligned} P_r^n &= n \times (n-1) \times \dots \times (n-r+1) \\ &= \frac{n \times (n-1) \times \dots \times (n-(r-1))(n-r)!}{(n-r)!} \\ &= \frac{n!}{(n-r)!}. \end{aligned}$$



## Example 6.

Four names are drawn from among the 24 members of a club for the offices of president, vice president, treasurer, and secretary. In how many different ways can this be done?

### Solution:

- The number of permutations of 24 distinct objects taken four at a time is

$$P_4^{24} = \frac{24!}{20!} = 24 \times 23 \times 22 \times 21 = 255\,024$$

## Example 7.

In how many ways can a local chapter of the American Chemical Society schedule three speakers for three different meetings if they are all available on any of five possible dates?

### Solution:

- Since we must choose three of the five dates and the order in which they are chosen (assigned to the three speakers) matters, we get

$$P_3^5 = \frac{5!}{2!} = \frac{120}{2} = 60$$

### Remark 2.

We might also argue that the first speaker can be scheduled in five ways, the second speaker in four ways, and the third speaker in three ways, so that the answer is  $5 \times 4 \times 3 = 60$ .



- We shall now determine the number of permutations of a set of  $n$  objects when certain of the objects are **indistinguishable** from each other.
- To set this situation straight in our minds, consider the following example.

## Example 8.

How many different letter arrangements can be formed from the letters **PEPPER**?

## Solution:

- We first note that there are  $6!$  permutations of the letters  $P_1E_1P_2P_3E_2R$  when the  $3P$ 's and the  $2E$ 's are distinguished from each other.
- However, consider any one of these permutations—for instance,  $P_1P_2E_1P_3E_2R$ . If we now permute the  $P$ 's among themselves and the  $E$ 's among themselves, then the resultant arrangement would still be of the form  $PPEPER$ .
- That is, all  $3!2!$  permutations

$$\begin{array}{ll} P_1P_2E_1P_3E_2R & P_1P_2E_2P_3E_1R \\ P_1P_3E_1P_2E_2R & P_1P_3E_2P_2E_1R \\ P_2P_1E_1P_3E_2R & P_2P_1E_2P_3E_1R \\ P_2P_3E_1P_1E_2R & P_2P_3E_2P_1E_1R \\ P_3P_1E_1P_2E_2R & P_3P_1E_2P_2E_1R \\ P_3P_2E_1P_1E_2R & P_3P_2E_2P_1E_1R \end{array}$$

are of the form  $PPEPER$ .

- Hence, there are  $\frac{6!}{3!2!} = 60$  possible letter arrangements of the letters  $PEPPER$ .

## Theorem 5.

The number of permutations of  $n$  objects of which  $n_1$  are of one kind,  $n_2$  are of a second kind,  $\dots, n_k$  are of a  $k^{\text{th}}$  kind, and  $n_1 + n_2 + \dots + n_k = n$  is

$$\frac{n!}{n_1! \times n_2! \times \dots n_k!}$$

## Example 9.

In how many ways can two paintings by Monet, three paintings by Renoir, and two paintings by Degas be hung side by side on a museum wall if we do not distinguish between the paintings by the same artists?

### Solution:

- Substituting  $n = 7$ ,  $n_1 = 2$ ,  $n_2 = 3$ , and  $n_3 = 2$  into the formula of Theorem 5, we get

$$\frac{7!}{2! \times 3! \times 2!} = 210.$$

## Example 10.

A chess tournament has 10 competitors, of which 4 are Russian, 3 are from the United States, 2 are from Great Britain, and 1 is from Brazil. If the tournament result lists just the nationalities of the players in the order in which they placed, how many outcomes are possible?

**Solution:** There are

$$\frac{10!}{4! \times 3! \times 2! \times 1!} = 12600 \text{ possible outcomes.}$$

## Example 11.

How many different signals, each consisting of 9 flags hung in a line, can be made from a set of 4 white flags, 3 red flags, and 2 blue flags if all flags of the same color are identical?

**Solution:** There are

$$\frac{9!}{4! \times 3! \times 2!} = 1260 \text{ different signals.}$$

- There are many problems in which we are interested in determining the number of ways in which  $r$  objects can be selected from among  $n$  distinct objects without regard to the order in which they are selected.

## Definition 2 (Combinations).

A combination is a selection of  $r$  objects taken from  $n$  distinct objects without regard to the order of selection.

**Example 12.**

In how many different ways can a person gathering data for a market research organization select three of the 20 households living in a certain apartment complex?

**Solution:**

- If we care about the order in which the households are selected, the answer is

$$P_3^{20} = 20 \times 19 \times 18 = 6840$$

- but each set of three households would then be counted  $3! = 6$  times.
- If we do not care about the order in which the households are selected, there are only  $\frac{6840}{6} = 1140$  ways in which the person gathering the data can do his or her job.

- Actually **combination** means the same as **subset** and when we ask for the number of combinations of  $r$  objects selected from a set of  $n$  distinct objects, we are simply asking for the total number of subsets of  $r$  objects that can be selected from a set of  $n$  distinct objects.
- In general, there are  $r!$  permutations of the objects in a subset of  $r$  objects, so that the  $P_r^n$  permutations of  $r$  objects selected from a set of  $n$  distinct objects contain each subset  $r!$  times.
- Dividing  $P_r^n$  by  $r!$  and denoting the result by the symbol  $C_r^n$ , we thus have the following theorem.

## Theorem 6.

The number of combinations of  $n$  distinct objects taken  $r$  at a time is

$$C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!} \quad (2)$$

for  $r = 0, 1, 2, \dots, n$ .

## Remark 3.

$C_r^n = \binom{n}{r}$  represents the number of different groups of size  $r$  that could be selected from a set of  $n$  objects when the **order of selection is not considered relevant**.



**Example 13.**

In how many different ways can six tosses of a coin yield two heads and four tails?

**Solution:**

- This question is the same as asking for the number of ways in which we can select the two tosses on which heads is to occur.
- Therefore, applying Theorem 6, we find that the answer is

$$C_2^6 = \frac{6!}{2! \times 4!} = 15$$

**Remark 4.**

This result could also have been obtained by the rather tedious process of enumerating the various possibilities, *HHTTTT, THTHT, HTHTTT, ...*, where *H* stands for head and *T* for tail.

## Example 14.

How many different committees of two chemists and one physicist can be formed from the four chemists and three physicists on the faculty of a small college?

### Solution:

- Since two of four chemists can be selected in  $C_2^4 = \frac{4!}{2!2!} = 6$  ways
- and one of three physicists can be selected in  $C_1^3 = \frac{3!}{1! \times 2!} = 3$  ways.
- Theorem 1 shows that the number of committees is  $6 \times 3 = 18$ .

## *Remark 5.*

- A combination of  $r$  objects selected from a set of  $n$  distinct objects may be considered a **partition** of the  $n$  objects into two subsets containing, respectively, the  $r$  objects that are selected and the  $n - r$  objects that are left.
- Often, we are concerned with the more general problem of partitioning a set of  $n$  distinct objects into  $k$  subsets, which requires that each of the  $n$  objects must belong to one and only one of the subsets.
- The order of the objects within a subset is of **no importance**.

**Theorem 7.**

The number of ways in which a set of  $n$  distinct objects can be partitioned into  $k$  subsets with  $n_1$  objects in the first subset,  $n_2$  objects in the second subset, . . . , and  $n_k$  objects in the  $k^{\text{th}}$  subset is

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! \times n_2! \times \dots \times n_k!}, \quad n = n_1 + n_2 + \dots + n_k.$$

## Proof of Theorem 7.

- Since the  $n_1$  objects going into the first subset can be chosen in  $\binom{n}{n_1}$  ways,
- the  $n_2$  objects going into the second subset can then be chosen in  $\binom{n-n_1}{n_2}$  ways,
- the  $n_3$  objects going into the third subset can then be chosen in  $\binom{n-n_1-n_2}{n_3}$  ways, and so forth,
- it follows by Theorem 2 that the total number of partitions is

$$\begin{aligned}
 \binom{n}{n_1, n_2, \dots, n_k} &= \binom{n}{n_1} \times \binom{n-n_1}{n_2} \times \dots \times \binom{n-n_1-n_2-\dots-n_{k-1}}{n_k} \\
 &= \frac{n!}{n_1! \times (n-n_1)!} \times \frac{(n-n_1)!}{n_2! \times (n-n_1-n_2)!} \\
 &\quad \times \dots \times \frac{(n-n_1-n_2-\dots-n_{k-1})!}{n_k! \times 0!} \\
 &= \frac{n!}{n_1! \times n_2! \times \dots \times n_k!}.
 \end{aligned}$$



**Example 15.**

In how many ways can seven businessmen attending a convention be assigned to one triple and two double hotel rooms?

**Solution:**

- Substituting  $n = 7$ ,  $n_1 = 3$ ,  $n_2 = 2$ , and  $n_3 = 2$  into the formula of Theorem 7, we get

$$\binom{7}{3, 2, 2} = \frac{7!}{3! \times 2! \times 2!} = 210.$$

## Theorem 8.

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r \quad \text{for any positif integer } n.$$

## Definition 3 (Binomial Coefficients).

The coefficient of  $x^{n-r}y^r$  in the binomial expansion of  $(x + y)^n$  is called the **binomial coefficient**  $\binom{n}{r}$ .

- The calculation of binomial coefficients can often be simplified by making use of the three theorems that follow.

## Theorem 9.

For any positive integers  $n$  and  $r = 0, 1, 2, \dots, n$ ,

$$\binom{n}{r} = \binom{n}{n-r}.$$



## Proof of Theorem 9.

- We might argue that when we select a subset of  $r$  objects from a set of  $n$  distinct objects, we leave a subset of  $n - r$  objects;
- hence, there are as many ways of selecting  $r$  objects as there are ways of leaving (or selecting)  $n - r$  objects.
- To prove the theorem algebraically, we write

$$\begin{aligned}\binom{n}{n-r} &= \frac{n!}{(n-r)! [n - (n-r)]!} = \frac{n!}{(n-r)! r!} \\ &= \frac{n!}{r!(n-r)!} = \binom{n}{r}.\end{aligned}$$



## Example 16.

- Given  $\binom{4}{0} = 1$ , and  $\binom{4}{1} = 4$ , find  $\binom{4}{3}$ , and  $\binom{4}{4}$ .
- Given  $\binom{5}{1} = 5$ , and  $\binom{5}{2} = 10$ , find  $\binom{5}{3}$ , and  $\binom{5}{4}$ .

## Solution:

1.

$$\binom{4}{3} = \binom{4}{4-3} = \binom{4}{1} = 4, \quad \text{and} \quad \binom{4}{4} = \binom{4}{4-4} = \binom{4}{0} = 1.$$

2.

$$\binom{5}{3} = \binom{5}{5-2} = \binom{5}{2} = 10, \quad \binom{5}{4} = \binom{5}{5-4} = \binom{5}{1} = 5.$$

## Theorem 10.

For any positive integer  $n$  and  $r = 1, 2, \dots, n-1$ , we have

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}.$$

## Proof of Theorem 10.

- Substituting  $x = 1$  into  $(x + y)^n$ , and write

$$(1 + y)^n = (1 + y)(1 + y)^{n-1},$$

- Since the coefficient of  $y^r$  in  $(1 + y)^n$  is  $\binom{n}{r}$ ,
- and the coefficient of  $y^r$  in  $(1 + y)^{n-1} + y(1 + y)^{n-1}$  is the sum of the coefficient of  $y^r$  in  $(1 + y)^{n-1}$ , that is,  $\binom{n-1}{r}$ , and the coefficient of  $y^{r-1}$  in  $(1 + y)^{n-1}$ , that is,  $\binom{n-1}{r-1}$
- By equating these coefficients, we obtain

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1},$$

which completes the proof. □

## Theorem 11.

$$\sum_{r=0}^k \binom{m}{r} \binom{n}{k-r} = \binom{m+n}{k}.$$

## Proof of Theorem 11.

Using the same technique as in the proof of Theorem 10, by equating the coefficients of  $y^k$  in the expressions on both sides of the equation

$$(1+y)^{m+n} = (1+y)^m (1+y)^n.$$



## Exercise 6.

An assembler of electronic equipment has 20 integrated-circuit chips on her table, and she must solder three of them as part of a larger component. In how many ways can she choose the three chips for assembly?

## Solution 6.

Using Theorem 4, we obtain

$$P_3^{20} = \frac{20!}{17!} = 20 \times 19 \times 18 = 6840.$$

**Exercise 7.**

A lot of manufactured goods, presented for sampling inspection, contains 16 units. In how many ways can 4 of the 16 units be selected for inspection?

**Solution 7.**

According to Theorem 6,

$$C_4^{16} = \frac{16!}{4! \times 12!} = \frac{16 \times 15 \times 14 \times 13}{4 \times 3 \times 2 \times 1} = 1820.$$

## Exercise 8.

In a primary election, there are four candidates for mayor, five candidates for city treasurer, and two candidates for county attorney.

- (a). In how many ways can a voter mark his ballot for all three of these offices?
- (b). In how many ways can a person vote if he exercises his option of not voting for a candidate for any or all of these offices?

## Solution 8.

- (a). According to Theorem 2, we obtain  $4 \times 5 \times 2 = 40$  ways.
- (b). Again according to Theorem 2, we get  $5 \times 6 \times 3 = 90$  ways.



**Exercise 9.**

Expressing the binomial coefficients in terms of factorials and simplifying algebraically, show that

(a).

$$\binom{n}{r} = \frac{n-r+1}{r} \cdot \binom{n}{r-1}$$

(b).

$$\binom{n}{r} = \frac{n}{n-r} \cdot \binom{n-1}{r}$$

(c).

$$n \binom{n-1}{r} = (r+1) \binom{n}{r+1}$$

**Solution 9.**

- (a). By using the binomial coefficient definition on the right hand of the equality, we get

$$\begin{aligned}
 \frac{n-r+1}{r} \cdot \binom{n}{r-1} &= \frac{n-r+1}{r} \times \frac{n!}{(r-1)!(n-(r-1))!} \\
 &= \frac{n-r+1}{r} \times \frac{n!}{(r-1)!(n-r+1)!} \\
 &= \frac{\cancel{n-r+1}}{r} \times \frac{n!}{(r-1)!(n-r)!\cancel{(n-r+1)}} \\
 &= \frac{n!}{r!(n-r)!} \\
 &= \binom{n}{r}.
 \end{aligned}$$

- (b). Use similar argument.  
 (c). Use similar argument.

## Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

## Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* [here](#).

## Some useful websites:

- Society of Actuaries, [Exam of probability, sample questions](#).
- Society of Actuaries, [Exam of probability, sample solutions](#).